

Maya Sankar

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Research Interests: Probabilistic combinatorics, extremal graph theory, topological methods in combinatorics

ACADEMIC POSITIONS

CLE Moore Instructor , Department of Mathematics, MIT	2026–present
Member , School of Mathematics, Institute for Advanced Study	2025–2026

EDUCATION

Stanford University Ph.D. in Mathematics, advised by Jacob Fox Thesis title: Topological Approaches to Extremal Combinatorics	2020–2025
Massachusetts Institute of Technology B.S. in Mathematics and Computer Science, Minor in Music	2016–2020

FELLOWSHIPS

Hertz Foundation Fellowship	2020–2025
NSF Graduate Fellowship	2020–2025
Stanford EDGE Doctoral Fellowship	2020–2025

PAPERS

- J. Balogh, H. Luo, and M. Sankar, A Jump in the Codegree Turán Densities of Long Tight Cycles, *International Mathematics Research Notices*, to appear. Available at [arXiv:2602.18398](#).
- N. Crawford, M. Sankar, S. Spiro, C. Schildkraut, Rainbow Trees in Hypercubes, *SIAM Journal of Discrete Mathematics*, to appear. Available at [arXiv:2508.14186](#).
- M. Sankar, The Turán density of 4-uniform tight cycles, *Advances in Combinatorics* (2026).
- M. Krebs and M. Sankar, Abelian groups without 3-chromatic Cayley graphs, *Journal of Combinatorial Theory, Series B* **179** (2026).
- M. Sankar, An improved Turán exponent for 2-complexes, *Random Structures & Algorithms* **68** (2026).
- J. Gao, S. Jiang, H. Liu, and M. Sankar, Generalized Ramsey–Turán density for cliques, *Forum of Mathematics, Sigma* **13** (2025).
- J. Fox, M. Sankar, M. Simkin, J. Tidor, and Y. Zhou, Ramsey and Turán numbers of sparse hypergraphs, preprint 2023. [arXiv:2401.00359](#)
- M. Sankar, The Turán number of surfaces, *Bulletin of the London Math Society* **56** (2024). [arXiv:2210.11041](#)
- M. Sankar, Homotopy and the homomorphism threshold of odd cycles, preprint 2022. [arXiv:2206.07525](#)

- M. Sankar, Further bijections to pattern-avoiding valid hook configurations, *Advances in Applied Mathematics* **125** (2021).

INVITED TALKS

Minisymposium on Probabilistic and Extremal Combinatorics, SIAM Conference on Discrete Mathematics 2026	Jun 2026
Flags in the Mountains, Colorado State University Mountain Campus	Jun 2026
Computer Science and Discrete Mathematics Seminar, Institute for Advanced Study	May 2026
Discrete Math Seminar, University of Massachusetts Amherst	Mar 2026
Richard P. Stanley Seminar in Combinatorics, MIT	Mar 2026
Discrete Math Seminar, Rutgers University	Feb 2026
Computer Science and Discrete Mathematics Seminar, Institute for Advanced Study	Dec 2025
Algorithms, Computation, and Optimization Seminar, Carnegie Mellon University	Dec 2025
Atlanta Lecture Series	Nov 2025
What Is...? Seminar, Institute for Advanced Study	Oct 2025
Discrete Math Seminar (Zoom), University of Victoria	Oct 2025
Discrete Mathematics Seminar, Princeton University	Oct 2025
Workshop in Combinatorics, Probability, and Computing, Oberwolfach	Sep 2025
Research Program Seminar, Park City Mathematics Institute	Jul 2025
Extremal Combinatorics Graduate Student Seminar Series, Simon–Laufer Mathematical Sciences Institute	Mar 2025
Special Sessions on Sparse Graphs: Colorings, Randomness, and Ramsey Theory and on Extremal Combinatorics and Random Discrete Structures, Joint Mathematics Meetings	Jan 2025
Combinatorics and TCS Seminar, University of Chicago	Jan 2025
Discrete Mathematics Seminar, Emory University	Nov 2024
Special Session on Topological and Geometric Combinatorics, AMS Fall Sectional Meeting	Oct 2024
Combinatorics Seminar, California Institute of Technology	Oct 2024
Discrete Mathematics Seminar, Princeton University	Sep 2024
Workshop on Synergies of Combinatorics and Theoretical Computer Science, EPFL	Aug 2024
Minisymposium on Recent Advances in Probabilistic and Extremal Combinatorics, SIAM Conference on Discrete Mathematics 2024	Jul 2024
Special Session on Extremal Combinatorics and Connections, AMS Spring Sectional Meeting	May 2024
9th Lake Michigan Workshop on Combinatorics and Graph Theory	Apr 2024
Graduate Student Combinatorics Conference	Mar 2024
Special Session on Bridging Applied and Quantitative Topology, Joint Mathematics Meetings	Jan 2024
European Conference on Combinatorics, Graph Theory, and Applications	Aug 2023
Random Structures and Algorithms Conference	Jun 2023
IAS Program for Women and Mathematics Evening Research Talks	May 2023
Special Session on Topological and Geometric Methods in Combinatorics, AMS Spring Sectional Meeting	Apr 2023
Combinatorics Seminar, MIT	Apr 2023

Discrete Math Seminar, Institute for Basic Science, Korea	Feb 2023
Virtual Discrete Math Colloquium, Institute for Basic Science, Korea	Dec 2022
Special Session on Extremal Graph Theory, AMS Fall Sectional Meeting	Oct 2022
SIAM Conference on Discrete Mathematics 2022	Jun 2022
Stanford Combinatorics Seminar	Apr 2022

PROFESSIONAL SERVICE

Reviewer for Combinatorics, Probability, and Computing; Forum of Mathematics, Sigma; Journal of Graph Theory; Journal of the London Mathematics Society.

TEACHING AND OUTREACH

Teaching Assistant, Stanford Math Department	
Math 61DM — Modern Mathematics: Discrete Methods	Fall 2022
Research Advisor, University of Minnesota Duluth REU	Summers 2023, 2024, 2025
Directed Reading Program Mentor, Stanford Math Department	2020, 2021, 2023, 2024
Stanford Women in Math Mentoring Program	
Mentor	2020–2025
Board Member	2021–2025
Teaching Assistant, Euler Circle	Summer 2021
PRIMES Circle Mentor, MIT Math Department	2018, 2019

UNDERGRADUATE RESEARCH EXPERIENCE

Combinatorics REU at University of Minnesota, Duluth	Summer 2019
MIT Undergraduate Research Opportunities Program	Fall 2016, Spring 2019